field-of-view telescopes, the teams were able to find nearly four dozen type Ia supernovae at various distances from earth. After painstakingly determining the distance and recessional velocities of each, both groups came to a totally unexpected conclusion: ever since the universe was about 7 billion years old, its expansion rate has *not* been decelerating. Instead, the expansion rate has been *speeding up.*

The groups concluded that the expansion of the universe slowed down for the first 7 billion years after the initial outward burst, much like a car slowing down as it approaches a highway tollbooth. This was as expected. But the data revealed that, like a driver who hits the gas pedal after gliding through the EZ-Pass lane, the expansion of the universe has been accelerating ever since. The expansion rate of space 7 billion years ATB was *less* than the expansion rate 8 billion years ATB, which was *less* than the expansion rate 9 billion years ATB, and so on, all of which are *less* than the expansion rate today. The expected deceleration of spatial expansion has turned out to be an unexpected *acceleration*.

But how could this be? Well, the answer provides the corroborating second opinion regarding the missing 70 percent of mass/energy that physicists had been seeking.

The Missing 70 Percent

If you cast your mind back to 1917 and Einstein's introduction of a cosmological constant, you have enough information to suggest how it might be that the universe is accelerating. Ordinary matter and energy give rise to ordinary attractive gravity, which slows spatial expansion. But as the universe expands and things get increasingly spread out, this cosmic gravitational pull, while still acting to slow the expansion, gets weaker. And this sets us up for the new and unexpected twist. If the universe should have a cosmological constant—and if its magnitude should have just the right, small value—up until about 7 billion years ATB its gravitational repulsion would have been overwhelmed by the usual gravitational attraction of ordinary matter, yielding a net slowing of expansion, in keeping with the data. But then, as ordinary matter spread out and its gravitational pull diminished, the repulsive push of the cosmological constant (whose strength does not change as matter spreads